



Governing AI Systems That Act

Executive Brief 2026

Runtime metrology, observability, and control of artificial autonomy

Measure execution. Govern action. Produce evidence.

When AI can act, governance must intervene before the next action

Artificial intelligence systems are no longer limited to producing responses. They can call tools, interact with information systems, trigger workflows, modify or transmit data, coordinate other agents, and initiate actions with limited human supervision.

This development changes the nature of risk.

The question is no longer simply whether an AI system can produce an acceptable recommendation. It is now whether the system should be authorized to continue, decide, or act in its current state.

For executives, risk leaders, and operational teams, four questions are becoming central:

Who can interrupt the agent?

Which signals trigger escalation?

Which actions require human approval?

Can the organization demonstrate what was measured, decided, and executed?

A governance policy without runtime measurement remains an intention. To govern an AI system in action, an organization must be able to observe its behavior during execution, detect certain regime changes, and apply an appropriate policy before the next action is initiated.

Making governance operational

NeoMundi is developing an independent runtime metrology and observability infrastructure for generative and agentic artificial intelligence systems.

Its platform, ControlTower, measures the behavioral conditions under which a response, decision, or action is produced. It transforms these observations into signals that can be used by the organization's systems.

These signals can support authorization, escalation, regeneration, interruption, or human-validation logic. They can also be associated with timestamped artifacts documenting what occurred during execution.

NeoMundi does not replace the organization's governance policy. The platform does not claim to determine by itself whether a response is true, acceptable, or authorized.

NeoMundi measures.

Your policy decides.

Your teams retain the evidence.

The previous brief already established this separation between the measurement instrument and decision-making authority. This new version extends it from generated responses to systems capable of acting.

From runtime signals to governance decisions

The operational principle is straightforward.

An AI system or agent produces a response, an intermediate decision, or an intended action. ControlTower then observes its behavioral state, either after generation or directly during execution when the level of criticality requires it.

The platform produces metrics and signals relating in particular to stability, coherence, semantic variation, the evolution of the generation regime, and the quality of the available trace.

These signals are then interpreted according to the policy defined by the organization. Depending on the business context, the system may be authorized to continue, routed toward regeneration, transferred to another model, submitted for human validation, interrupted, or simply documented.

Measurement does not therefore constitute an additional autonomous decision. It provides the information required to apply the governance decision at the appropriate time.

AI system → runtime measurement → signal → internal policy → governed action

Targeting human oversight

The governance of agentic AI cannot rely on having a human permanently monitoring every execution.

Undifferentiated supervision would be costly, slow, and difficult to sustain at scale. Conversely, full autonomy without instrumentation would make atypical behaviors difficult to detect and decisions difficult to reconstruct.

The objective is therefore to make human intervention more targeted, faster, and proportionate to the level of risk.

ControlTower helps identify the executions that require heightened attention and distinguish situations that can be handled automatically from those that should be escalated.

**Continuous observation does not mean blocking everything.
It means intervening in the right cases, at the right time.**

This approach allows the organization to reserve human expertise for situations in which it creates genuine value: high ambiguity, sensitive actions, regime changes, threshold breaches, insufficient evidence, or an inability to reverse the effects of an action after execution.

Measurable operational value

NeoMundi does not present runtime governance as an abstract promise. Its value must be measured within the organization's actual operating environment.

A pilot establishes a baseline of current behavior, identifies situations requiring supervision, and evaluates the effect of a governance policy on cost, time, and risk.

The first dimension measured is the human oversight burden. The aim is to determine what proportion of executions genuinely requires review, rather than indiscriminately checking every output.

The second dimension concerns response speed. An organization can measure the time between the emergence of a signal, its interpretation, and effective intervention by a system or operator.

The third dimension concerns prevention. It becomes possible to count actions that were interrupted, regenerated, or escalated before producing a difficult-to-reverse effect.

Finally, the platform enables the organization to assess the quality of traceability: the proportion of executions supported by a complete record, the ability to reconstruct a decision, signal coverage, and the availability of the artifacts required for audit.

Possible pilot indicators

Executive concern	Measurable indicator
Human oversight	Share of executions actually escalated
Responsiveness	Time between signal emergence and intervention

Executive concern	Measurable indicator
Operational risk	Number of actions intercepted before execution
Cost	Cost of review, regeneration, and incident handling
Auditability	Share of executions supported by a complete trace
Governance	Distribution across authorization, escalation, correction, and interruption
Quality of service	Effect of governance on latency and throughput
Provider strategy	Behavioral differences across models, versions, or architectures

The ROI question can therefore be addressed concretely:

What is the current cost of untargeted supervision, late detection, or insufficiently documented actions?

And then:

What proportion of this cost can be reduced through runtime measurement and an appropriate intervention policy?

Two levels of governance

ControlTower adapts to the system's level of criticality, architecture, and action reversibility.

OBS — Governance observability

OBS is the natural entry point for systems whose responses can still be reviewed, corrected, moderated, or regenerated after production.

In this mode, prompts and responses remain within the organization's environment. NeoMundi receives only the normalized metrics and technical artifacts required for observation.

Never the prompt. Never the response.

OBS makes it possible to establish a behavioral baseline, monitor changes over time, compare models or providers, and identify executions requiring further analysis.

This mode is particularly suitable for internal assistants, business copilots, content-generation systems, document-retrieval applications, and workflows in which correction remains possible before a definitive operational effect occurs.

OBS also allows the organization to progressively develop its governance policy. It can begin by observing real systems, understanding the distribution of signals, and determining which events justify escalation.

GOV — Governance during execution

GOV is designed for systems whose decisions or actions are difficult to correct after execution.

In this mode, ControlTower operates within the execution path and measures how behavior evolves during generation. The resulting signals can be used before the agent proceeds to its next action.

Depending on the chosen policy, the system may continue normally, temporarily interrupt execution, regenerate its response, consult another model, restrict its capabilities, request human approval, or block the proposed action.

Content passes through temporarily when required for runtime orchestration. It is not retained, logged, or stored by ControlTower NeoMundi.

GOV is particularly relevant for autonomous agents, multi-agent systems, financial or fiduciary operations, critical infrastructure, sensitive legal workflows, healthcare systems, and processes capable of acting directly on third-party tools or environments.

The determining criterion: reversibility

The choice between OBS and GOV does not depend solely on the industry or the power of the model being used.

The main criterion is simpler:

Can an incorrect output or inappropriate action still be corrected after execution?

When the answer is yes, observation outside the critical execution path, combined with moderation, regeneration, or downstream human review, may be sufficient. OBS is then generally the most rational starting point.

When the answer is no, the organization must be able to intervene before the next action. This is particularly important when a decision commits funds, modifies infrastructure, transmits sensitive advice, triggers an operation, or produces an effect that is difficult to reverse.

In these situations, runtime governance becomes relevant.

Measure what can be recovered.

Control what cannot.

Provider-independent metrology

Traditional monitoring tools mainly observe availability, latency, cost, technical errors, and final outputs.

These dimensions remain essential. However, they are not sufficient to characterize the behavior of a generative or agentic system during execution.

A response may be fluent, plausible, and technically valid while having been produced under a behavioral regime that warrants heightened attention.

Conversely, behavioral variation does not automatically mean that the response is incorrect.

A signal is not a verdict.

NeoMundi observes complementary dimensions: stability, coherence, semantic variation, regime evolution, measurement coverage, risk signals, and trace completeness.

The purpose is not to replace business evaluation or human judgment. It is to make behavioral phenomena visible when they would remain difficult to observe through an analysis of the final output alone.

This independent layer also allows organizations to compare behavior over time, across providers, model versions, or agent architectures.

Producing evidence, not only alerts

When an incident occurs, it is not enough to know that an alert was issued.

The organization must be able to reconstruct the context of execution. It must determine which system was observed, when it was observed, under which configuration, which signals were produced, which policy was applied, and which decision was ultimately made.

ControlTower can associate each observation with an identifier, timestamp, metrics, governance signals, a processing decision, and configurable audit artifacts.

Depending on the architecture selected, this trace can also document the human or technical authority able to intervene, the escalation conditions, and the elements required to maintain a chain of custody.

The objective is not to ask teams, auditors, or regulators to trust a promise.

The objective is to provide usable evidence.

From declared compliance to operational control

ControlTower does not replace legal analysis, certification, or a comprehensive risk-management policy.

However, the platform provides several technical capabilities that directly support governance and compliance efforts: continuous monitoring, behavioral observability, execution traceability, signals for human oversight, incident documentation, and longitudinal comparison.

It helps the organization connect its governance requirements with what actually occurs during operation.

Compliance describes what should be done.

Metrology makes it possible to observe what is actually happening.

This distinction becomes essential when systems evolve, providers update their models, or behavior changes without any visible modification to the business application.

A privacy-first architecture

ControlTower is designed around the principle of data minimization.

In OBS mode, no prompt or response is transmitted to NeoMundi. Only the artifacts and metrics required for measurement are processed.

In GOV mode, content may pass through temporarily when required for runtime orchestration. This content is not retained, logged, or stored by NeoMundi.

Measurement, decision, and traceability artifacts remain configurable according to the required level of evidence, the organization's obligations, and its security architecture.

Architectural principles

Zero content retention · Zero content logging · Zero content storage

A sovereign deployment may also be considered when infrastructure, data-location, or operational-control requirements demand it.

Starting with a measurable-value pilot

The recommended starting point is not a system-wide deployment.

It is the selection of a workflow in which the current lack of observability already creates an identifiable cost, risk, or supervision burden.

This workflow may involve an autonomous agent, a business assistant, a multi-model system, or a process whose decisions must be documented.

The pilot begins with an observation phase. NeoMundi establishes a baseline of existing behavior without immediately modifying the production policy.

The signals are then analyzed to identify situations in which intervention, review, or escalation would be valuable.

Based on these observations, the organization can define an initial governance policy. This policy specifies the conditions under which the system continues, regenerates, escalates, requests approval, or interrupts an action.

The final stage consists of measuring the effect of this policy on human workload, processing time, trace quality, and risk exposure.

One targeted workflow.

One baseline.

One measurable policy.

One documented decision.

NeoMundi Observatory

NeoMundi is developing a global program for the longitudinal observation of artificial intelligence system behavior.

The objective is to create an independent infrastructure capable of observing system changes over time, identifying silent regime shifts, and producing reproducible and auditable evidence.

The Observatory currently analyzes a de-identified panel of AI providers. It publishes barometers, behavioral maps, and methodological protocols designed to advance the metrology of artificial intelligence systems.

Providers remain anonymized. The Observatory does not produce commercial rankings and does not seek to designate a winning model.

Its function is different:

To make visible behaviors that would otherwise remain unmeasured.

Longitudinal observation makes it possible to distinguish an isolated incident from a persistent change, track certain regime shifts, and document the limitations of the methods used.

It also provides an applied-research foundation for the gradual improvement of runtime governance policies.

Development within an international technology ecosystem

NeoMundi is a member of **NVIDIA Inception**, a program supporting startups developing advanced technologies in artificial intelligence.

This membership supports NeoMundi's technological development and its integration into the international AI ecosystem.

The NeoMundi Observatory program is also being developed with the support of **Infomaniak**, a partner of the Observatory and a Swiss cloud provider committed to digital sovereignty, privacy, and infrastructure control.

This cooperation contributes to the development of an observation infrastructure aligned with NeoMundi's principles of data minimization, independence, and traceability.



infomaniak
The Ethical Cloud 

Governing autonomy

The adoption of agentic AI is shifting the center of gravity of governance.

It is no longer sufficient to evaluate a response after it has been produced. Organizations must be able to measure the state of the system during execution, define the conditions under which it may continue or act, and retain evidence of the controls that were applied.

NeoMundi provides the metrology and observability layer required to make this governance operational.

Measure what the AI does.

Govern what it is authorized to do.

Produce the evidence.

Test ControlTower

Start with one targeted workflow.

Establish a behavioral baseline.

Measure the situations requiring supervision.

Build a governance policy adapted to your level of risk.

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